

Coupled Multi-Agent Language Flows on a Frozen 1.8B Diffusion LM: A Control-Variant Diagnostic Report

Project **ELF-MAS** · Phase 3 · Dataset **synth_2fact_v1** · Backbone Cola-DLM (1.8B, frozen) · Seed 0 · 2026-06-03

This report documents the current experimental evidence for role-conditioned agent specialization in the Phase 3 Cola+MAS setup. It describes the synthetic diagnostic dataset and its provenance, the model/agent wiring, the decode-and-evaluate pipeline, the control variants, the lesion-matrix results obtained so far, and a known implementation bug (now fixed and guarded) that invalidated two earlier baselines.

1 Dataset Construction: **synth_2fact_v1**

synth_2fact_v1 is **not** a public or downloaded benchmark. It is a **procedurally generated synthetic diagnostic dataset**, built specifically so that the correct per-agent lesion behavior is known by construction.

Provenance.

- Generator: `src/elf_mas/data/synth_2fact.py`.
- Design specification: `notes/synthetic_task_v1.md`.
- Generated deterministically with `numpy.random.default_rng(seed=42)`.
- Saved as a HuggingFace Dataset under `data/synth_2fact_v1/{train,eval}`.

Closed-world knowledge graph. A small fixed vocabulary — 100 people, 50 cities, 20 countries, 40 occupations, 20 hobbies — combined through five relation templates (e.g. `lives_in`: “X lives in Y”; `city_in`: “Y is in X”; `capital_of`, `works_as`, `hobby_of`).

Item contents. Each item contains a question Q , a private context ctx_A (given only to Agent A), a private context ctx_B (given only to Agent B), and a short gold answer. Each context carries one query-relevant fact (when applicable) plus 2–3 distractor facts of the same schema, so the model must use its private context conditionally rather than copy.

Bucket	Question Q	ctx_A relevant fact	ctx_B relevant fact	Gold	Why this bucket
AB	“What country is the city where Alice lives in?”	“Alice lives in Berlin.”	“Berlin is in Germany.”	Germany	Needs <code>lives_in</code> (A) composed with <code>city_in</code> (B)
A-only	“What does Alice do for work?”	“Alice works as a pilot.”	(distractors only)	pilot	Answer fact lives only in A
B-only	“What country is Cairo in?”	(distractors only)	“Cairo is in Egypt.”	Egypt	Answer fact lives only in B
neither	“What hobby does Charlie enjoy?”	(no relevant fact)	(no relevant fact)	unknown	Unanswerable; tests confabulation suppression

Table 1: Concrete items, one per bucket. Bucket membership is the ground truth for which agent’s private context is strictly necessary.

2 Dataset Distribution and Bucket Semantics

Every item belongs to exactly one of four buckets, defined by which private context is required to answer Q : **AB** (needs both, a two-hop composition), **A-only** (needs A), **B-only** (needs B), and **neither** (unanswerable from either).

	AB	A-only	B-only	neither
Intended proportion	40%	25%	25%	10%
Observed eval subset ($n=200$)	82 (41.0%)	48 (24.0%)	53 (26.5%)	17 (8.5%)

Table 2: Intended bucket proportions vs. the observed distribution of the 200-item eval subset used for the lesion matrix in §7. The AB (two-hop) bucket is the headline bucket ($n=82$, $\approx 41\%$). Full splits are train 50k / eval 5k.

Why the dataset is useful. Because the necessary context per item is known by construction, the dataset yields a *ground-truth lesion expectation*:

- Removing Agent A should hurt **AB** and **A-only**.
- Removing Agent B should hurt **AB** and **B-only**.
- Removing both should collapse **AB/A-only/B-only**, while **neither** stays at the “unknown” floor.

This makes lesion specificity directly checkable. The task is a **controlled synthetic diagnostic**, not a standard public multi-agent benchmark; it is intended to test the mechanism, not to claim downstream performance.

3 Model and Agent Connection

Frozen backbone. Cola-DLM: a DiT (1829.9M params) and a VAE (501.9M params), both frozen throughout. **MAS injection.** Multi-agent structure is added as in-block, LoRA-style per-agent cross-attention at DiT layers $\{8, 12, 16, 20\}$ (inner dim 512, 8 heads). **Two-agent structure.** Agent A receives ctx_A and Agent B receives ctx_B ; the question Q is supplied as prefix conditioning. Private contexts enter the model *only* through the MAS heads (not the prefix). **Trainable residual path.** Each agent’s cross-attention produces a velocity residual that is gated and added into the frozen backbone’s hidden state; the gate bias is initialized at -2 (initial gate ≈ 0.12) and the residual output projection is zero-initialized (untrained injection is a no-op). Total trainable: $\approx 25.3\text{M}$ params (dual configuration).

Stage	Role
Question Q	prefix conditioning (KV-cache priming; frozen backbone)
$\text{ctx}_A \rightarrow$ Agent A	private context, enters only via MAS cross-attn head A
$\text{ctx}_B \rightarrow$ Agent B	private context, enters only via MAS cross-attn head B
MAS residual injection	gated per-agent velocity residuals added at DiT layers $\{8, 12, 16, 20\}$
Frozen Cola-DLM DiT	produces velocity v over the latent answer block
Frozen VAE decoder	decodes predicted clean latents to answer tokens (§5)

Table 3: Data flow. Only the per-agent cross-attention heads and gates are trainable; the DiT and VAE are frozen.

4 Decoder / Evaluation Pipeline

Generation. The answer block is produced by rectified-flow denoising of a latent sequence: the DiT (with MAS residuals) predicts a velocity v , integrated by Euler steps from noise toward the clean latent. **Decoding via the frozen VAE.** The clean-latent estimate uses the x -prediction shortcut $z_{\text{clean}} = z_t - t v$, which is decoded to tokens through the frozen Cola VAE. **Auxiliary loss.** During training a token cross-entropy loss ($\lambda_{\text{CE}}=0.3$) is applied to these VAE-decoded tokens, in addition to the velocity MSE; it pressures decode-correct (not merely type-correct) latents.

Primary metric: per-token-ID exact match. The gold answer is tokenized and the first K predicted token IDs are compared directly. **Text-level EM is not used as the primary metric:** a decode-strip artifact joins the answer token to trailing pad-region tokens (e.g. "Belgium" + trailing region $\rightarrow$ "BelgiumTor"), so text-EM can read 0% even when the first answer token is correct.

Inference settings. $T_{\text{inf}}=16$ Euler steps; prefix-conditioned with KV-cache priming; the lesion matrix is computed by ablating gates/contexts at sample time (AB both agents active; A_only keep A; B_only keep B; Neither both removed).

5 Variant Definitions

ID	Name	Type	Purpose
B0	<code>full</code>	—	Two role-conditioned agents with private ctx_A , ctx_B (the model under test).
B6	<code>identical_ctx</code>	data-side	Both agents receive the same context: tests whether <i>distinct</i> private contexts matter.
B7	<code>context_shuffled</code>	data-side	Contexts permuted across examples: breaks the context-question correspondence.
B4	<code>single_model</code>	architecture-side	One parameter-matched head over $\text{concat}(\text{ctx}_A, \text{ctx}_B)$: isolates coupling from raw capacity.
B2	<code>frozen_agents</code>	architecture-side	Dual architecture with agent transforms frozen (gates-only training): tests whether trained agent transforms are necessary.

Table 4: Variants. Data-side variants change only the context fed to the model; architecture-side variants change the MAS wiring or which parameters train.

6 Results: Lesion Matrix

Variant	Full EM	rmA (B_only)	rmB (A_only)	rm_both (Neither)	Δ_a	Δ_b	Interpretation
B0 full	0.585	0.159	0.000	0.000	+0.427	+0.585	Both agents causally load-bearing; B more critical than A
B4 single_model	0.317	0.305	0.159	0.000	+0.012	+0.159	Param-matched single head; $\approx$ half of B0; barely uses A’s context
B6 identical_ctx	0.256	0.049	0.012	0.000	+0.207	+0.244	EM more than halves; lesion asymmetry compresses
B7 context_shuffled	0.061	0.012	0.012	0.000	+0.049	+0.049	Near-collapse; lesions small and symmetric
B2 frozen_agents	0.000	0.000	0.000	0.000	+0.000	+0.000	No-MAS floor; trained agent residuals necessary

Table 5: AB (two-hop) bucket, per-token EM (fraction in $[0, 1]$), single seed (seed 0); rows ordered by Full EM. $\Delta_a = \text{EM}(\text{full}) - \text{EM}(\text{remove A})$; $\Delta_b = \text{EM}(\text{full}) - \text{EM}(\text{remove B})$. **rmA/B_only** keeps only Agent B; **rmB/A_only** keeps only Agent A. All five variants are now valid and trained from distinct checkpoints (§7). The B0 row reproduces a prior standalone 30k run (0.585) to the reported precision.

Data-side controls. The monotone ordering $\text{B0 } 0.585 \gg \text{B6 } 0.256 \gg \text{B7 } 0.061$, with the lesion asymmetry strongest in B0 ($\Delta_b = +0.585 > \Delta_a = +0.427$), compressing under B6, and collapsing to near-symmetric/near-zero under B7, is consistent with the ground-truth lesion expectation of §3 and supports that performance is driven by role-distinct, correctly-paired private contexts rather than the two-head architecture alone (graded: B6 retains partial signal).

Architecture-side controls. At matched parameters ($\approx 25.3\text{M}$ vs. 25.5M), B0 (0.585) nearly doubles the single-head B4 (0.317), which supports that the benefit comes from coupled role-conditioning rather than raw capacity; B4’s lopsided lesions ($\Delta_a = +0.012$, $\Delta_b = +0.159$) indicate the single head leans on B’s context and barely integrates A’s—the failure mode expected on a two-hop bucket. B2 (**frozen_agents**) collapses to 0.000 (highest training loss, 3.82), supporting that *trained* agent residuals—not gates alone—are necessary. These conclusions are single-seed (see Next Steps).

7 Implementation Bug and Validity Notes

Silent fallback bug. In `--lora_mode`, the variants `frozen_agents` (B2) and `single_model` (B4) previously built the same dual `full` model: the LoRA branch never branched on `--variant`. The prior B0/B2/B4 checkpoints were therefore byte-identical (md5 4c64d210..., identical final loss 1.7658), and those B2/B4 results are invalid. **The earlier B2/B4 results have been discarded and must not be cited.** The valid variants (B0/B6/B7) are unaffected because their behavior is realized on the data side. (Separately, an initial parse transposed the eval-JSON axes, giving an incorrect B0 `rmA`= 0.358; the corrected value is 0.159, used above.)

Guard added. A runtime assertion (`assert_lora_variant_realized`) now fails loudly unless the requested variant actually reshaped the model: `single_model` must build single-mode wrappers and be parameter-matched ($0.85\text{--}1.15\times$ the dual reference); `frozen_agents` must have $< 10\%$ of MAS parameters trainable; `full/identical_ctx/context_shuffled` must train 100%. The eval path also warns when `--variant` disagrees with the checkpoint’s recorded variant.

8 Current Results and Next Steps

Variant	Status	Note
B0 <code>full</code>	completed, valid	headline 0.585 EM
B4 <code>single_model</code>	completed, valid	0.317; param-matched capacity control
B6 <code>identical_ctx</code>	completed, valid	0.256; data-side control
B7 <code>context_shuffled</code>	completed, valid	0.061; data-side control
B2 <code>frozen_agents</code>	completed, valid	0.000; no-MAS floor
seeds 1–2	not yet run	all variants single-seed so far
B2/B4 (prior sweep)	invalid, discarded	silent fallback to <code>full</code> (§7)

Table 6: Run status. All current results are single-seed (seed 0).

Conclusion. All five variants are now valid. On this controlled synthetic diagnostic task the results support role-conditioned specialization: (i) the AB-bucket EM and its lesion asymmetry degrade monotonically as role-relevant input structure is removed ($B0 \gg B6 \gg B7$), matching the by-construction lesion expectation; (ii) at matched parameters the coupled model nearly doubles a single concat-context head (B0 0.585 vs. B4 0.317), supporting that the gain is from coupling rather than capacity; and (iii) freezing the agent transforms collapses performance to the no-MAS floor (B2 0.000), supporting that trained agent residuals are necessary. All results are single-seed, so effect *magnitudes* are not yet bounded by variance, though the qualitative ordering is consistent across four independent controls.

Next steps. (i) Add seeds 1–2 for all five variants to obtain error bars (idle GPUs currently permit parallel execution). (ii) Commit the dataset generator (`src/elf_mas/data/synth_2fact.py`) for reproducibility — it is currently present only on the compute server and not tracked in the repository.